

Linear Regression with Bike Sharing Data

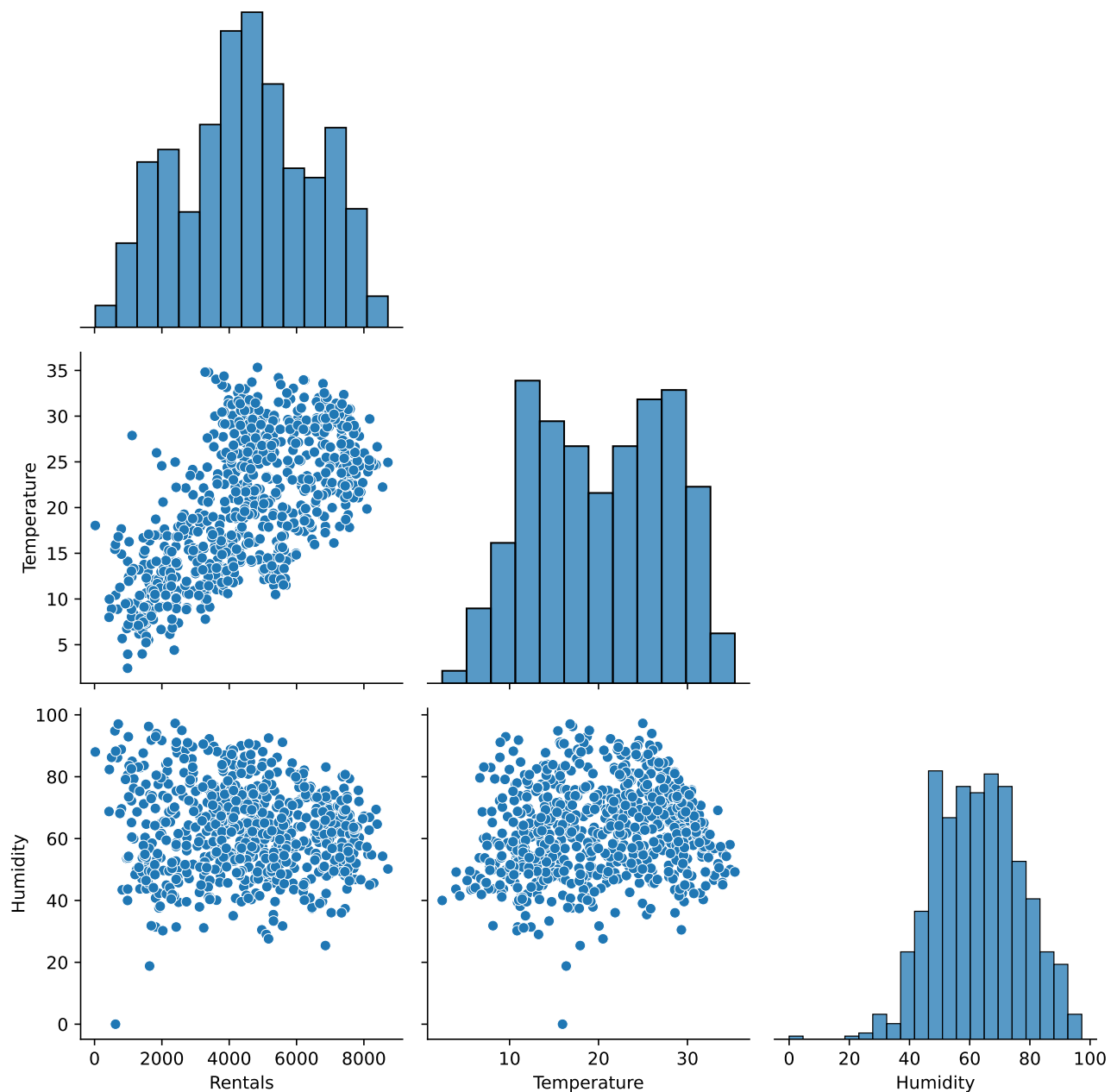
A city operates a bike-sharing system. We want to understand what affects the number of bikes rented on a given day and use regression to predict demand.

The dataset contains the following variables:

- **rentals**: total number of bike rentals that day
- **temperature**: temperature in degrees Celsius
- **hum**: humidity in percentage
- **season**: season of the year

1 Exploratory analysis

The following scatterplot shows the relationship between the number of bike rentals and the other variables included in the dataset.



Describe the relationship between the different variables in the dataset.

1.1 Which variables are related to each other?

1.2 Which variables appear to influence the number of bike rented?

2 Simple linear regression

We fit a model using only temperature as covariate:

$$\text{rentals} = \beta_0 + \beta_1 \text{temperature} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 1)$$

OLS Regression Results

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Dep. Variable:          rentals    R-squared:                0.394
Model:                  OLS        Adj. R-squared:           0.393
Method:                 Least Squares    F-statistic:             473.5
Date:                  Thu, 17 Sep 2026    Prob (F-statistic):       2.81e-81
Time:                  14:45:42    Log-Likelihood:          -6386.8
No. Observations:      731        AIC:                     1.278e+04
Df Residuals:          729        BIC:                     1.279e+04
Df Model:              1
Covariance Type:       nonrobust
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              coef      std err          t      P>|t|      [0.025      0.975]
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Intercept    1214.6421    161.164      7.537      0.000     898.242    1531.042
temp         161.9685      7.444     21.759      0.000     147.355     176.582
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Omnibus:                 20.477    Durbin-Watson:           0.468
Prob(Omnibus):           0.000    Jarque-Bera (JB):        12.566
Skew:                    0.167    Prob(JB):                0.00187
Kurtosis:                2.452    Cond. No.                62.6
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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

2.1 Interpret the temperature coefficient.

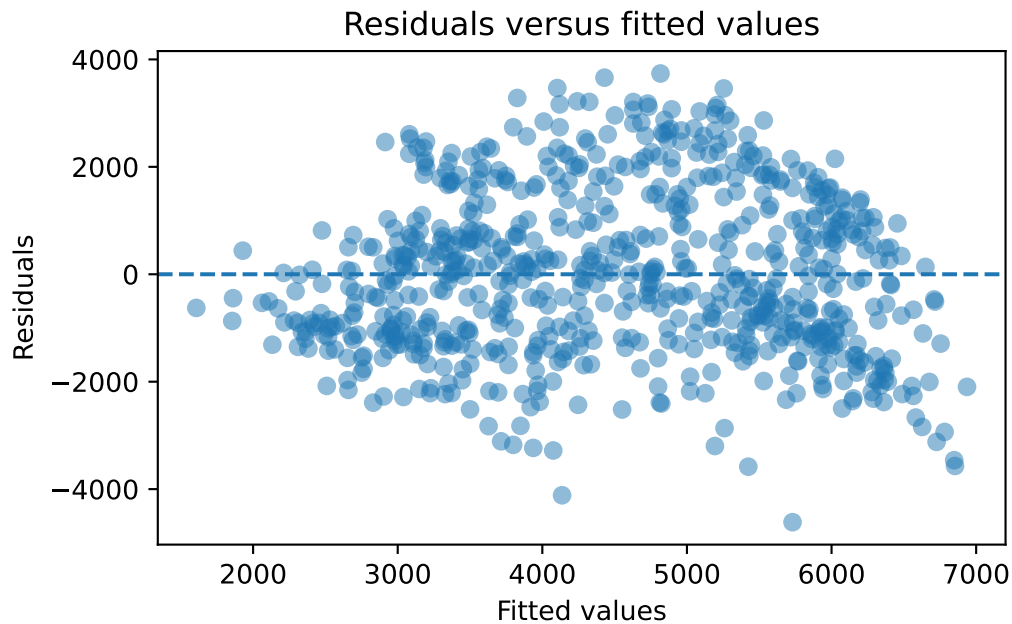
2.2 What does $R^2 = 0.39$ mean?

2.3 Calculate the predicted number of rentals for a day with temperature 20°C .

2.4 In one day with temperature 20°C we observe that 2927 bikes are rented, what is the residual? how do you interpret the sign?

2.5 Does the small p-value tell us that temperature is important in a practical sense? Explain the distinction between statistical significance and practical importance.

2.6 Below you see a plot of the model residuals against fitted values. What does this residual plot suggest about the fitted linear regression model?



2.7 Sketch a residual plot that would be consistent with a well-behaved linear regression model.

3 Multiple regression

We now include temperature and humidity and season to the model

$$\text{rentals} = \text{temperature} + \text{humidity} + \text{season} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$$

The fitted model is

OLS Regression Results

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Dep. Variable:          rentals    R-squared:                0.510
Model:                  OLS       Adj. R-squared:           0.506
Method:                 Least Squares    F-statistic:              150.8
Date:                   Thu, 17 Sep 2026    Prob (F-statistic):       1.11e-109
Time:                   14:45:42    Log-Likelihood:           -6309.1
No. Observations:       731    AIC:                      1.263e+04
Df Residuals:           725    BIC:                      1.266e+04
Df Model:                5
Covariance Type:        nonrobust
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               coef      std err          t      P>|t|      [0.025      0.975]
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Intercept      2500.8267    265.044      9.436      0.000     1980.482     3021.171
C(season)[T.Summer]  874.4701    187.194      4.671      0.000      506.962     1241.978
C(season)[T.Fall]   460.0173    246.006      1.870      0.062     -22.951      942.986
C(season)[T.Winter] 1562.4827    158.231      9.875      0.000     1251.837     1873.128
temp             163.8560     12.072     13.573      0.000      140.155      187.557
hum              -32.5439      3.641     -8.939      0.000     -39.691      -
25.396
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Omnibus:            5.118    Durbin-Watson:           0.468
Prob(Omnibus):      0.077    Jarque-Bera (JB):         3.843
Skew:               0.033    Prob(JB):                 0.146
Kurtosis:           2.651    Cond. No.                  439.
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Notes:

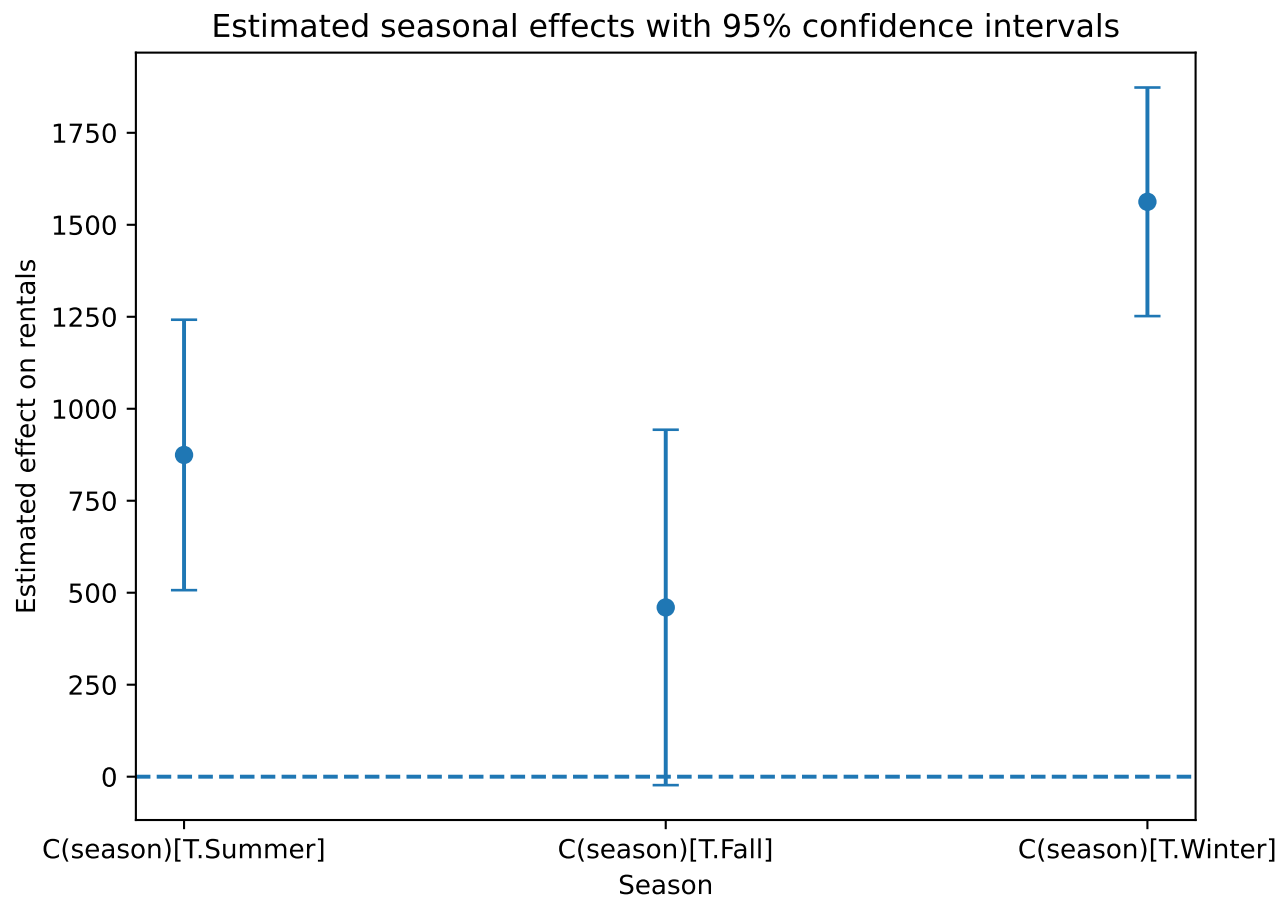
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

3.1 We included season as a categorical covariate. Why this choice?

3.2 Why in the output there are only 3 parameters for season?

3.3 Draw two fitted regression lines, one for humidity = 20, season = spring and temperature varying from 0 to 30 and the other for humidity = 20, season = winter and temperature varying from 0 to 30 and the other

3.4 Below are the estimated effect of season. How do you interpret them?



3.5 “One degree more of humidity causes a decrease of ca 32 in the number of rented bike”. Is this interpretation correct? How would you interpret the coefficient for humidity?

3.6 R^2 increased from 0.39 to 0.51 when additional predictors were included. What does this tell us?

4 Model with interactions

OLS Regression Results

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Dep. Variable:          rentals    R-squared:                0.550
Model:                  OLS        Adj. R-squared:           0.545
Method:                 Least Squares    F-statistic:              110.1
Date:                   Thu, 17 Sep 2026    Prob (F-statistic):       1.21e-119
Time:                   14:45:42    Log-Likelihood:           -6278.2
No. Observations:       731        AIC:                      1.257e+04
Df Residuals:           722        BIC:                      1.262e+04
Df Model:               8
Covariance Type:        nonrobust

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	coef	std err	t	P> t	[0.025	0.975]
Intercept	1901.4789	352.867	5.389	0.000	1208.712	2594.246
C(season) [T.Summer]	1602.4697	531.562	3.015	0.003	558.878	2646.061
C(season) [T.Fall]	8388.2465	1023.314	8.197	0.000	6379.221	1.04e+04
C(season) [T.Winter]	2023.7786	497.545	4.068	0.000	1046.971	3000.587
hum	-38.5183	3.597	-10.709	0.000	-45.580	-31.457
temp	241.4794	23.202	10.408	0.000	195.928	287.031
temp:C(season) [T.Summer]	-66.6060	30.079	-2.214	0.027	-125.658	-7.554
temp:C(season) [T.Fall]	-317.6347	40.895	-7.767	0.000	-397.922	-237.347
temp:C(season) [T.Winter]	-46.6199	32.152	-1.450	0.147	-109.742	16.502
Omnibus:	7.575	Durbin-Watson:	0.499			
Prob(Omnibus):	0.023	Jarque-Bera (JB):	5.666			
Skew:	-0.099	Prob(JB):	0.0588			
Kurtosis:	2.616	Cond. No.	1.47e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.47e+03. This might indicate that there are strong multicollinearity or other numerical problems.

4.1 What is the estimated coefficient of the temperature in Spring and Fall? and what is the estimated intercept?

4.2 Below is a plot that shows estimated regression lines, with confidence and prediction intervals, for Spring and Fall

- Individuate which lines is for Spring and which one is for Fall
- Explain the difference between prediction and confidence intervals
- Why is one type of interval so much wider than the other one?

